

MA1001 — Calculus for Science Part 1

Winter 2022 Examination · Paper 1 · Full Marking Scheme

Module	MA1001 — Calculus for Science Part 1
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Institution	University College Cork
Session	Winter 2022, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer 4 of the 5 questions . All questions carry equal marks (17.5 each, 70 in total). Marks for each part are shown in brackets.
Materials	No calculators. Only the log tables provided with the paper may be used. Formula sheet supplied (calculus, trigonometry, length/area/volume).

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. On the day you answer 4 questions, but this scheme works through **all five** in full, so you can pick the ones you're strongest on.

No calculators. Every answer is exact (for example $\ln 2$, $\pi \text{ m}^3$, $1/6 \text{ cm/s}$) and all the arithmetic is shown by hand.

Source basis. Every answer is built **only** from the MA1001 lecture notes and example/tutorial sheets: derivative rules (product, quotient, chain, linear substitution), critical points, optimisation, related rates, velocity and acceleration, and implicit differentiation. No textbook or outside material is used, except where a highlighted note says the notes don't cover something (Q3(a)).

Mark splits. Part totals are the paper's own. The step-by-step split within each part is Study-Uni's suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Study-Uni

Question 1

Differentiation: product, quotient and chain rules, power rule — 17.5 marks

1(a)

[6 marks]

Differentiate $f(x) = 4 \sin(2x) \cdot \ln(8 - 4x^2)$.

ANSWER

$$f'(x) = 8 \cos(2x) \ln(8 - 4x^2) - \frac{32x \sin(2x)}{8 - 4x^2} = 8 \cos(2x) \ln(8 - 4x^2) - \frac{8x \sin(2x)}{2 - x^2}$$

RULES YOU NEED

- Product rule: if $y = uv$ then $dy/dx = (du/dx)v + u(dv/dx)$.
- Linear substitution: the derivative of $\sin(ax)$ is a $\cos(ax)$.
- Chain rule: the derivative of $\ln(w)$ is $(1/w) \times dw/dx$.

WORKING — STEP BY STEP

1 Spot the structure: a product

f is two functions multiplied together, so use the product rule. Name them:

$$u = 4 \sin(2x) \quad v = \ln(8 - 4x^2)$$

$$\frac{df}{dx} = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

2 Differentiate $u = 4 \sin(2x)$

$\sin$ of (a number $\times x$): by linear substitution, the derivative of $\sin(ax)$ is a $\cos(ax)$. Here $a = 2$, and the 4 stays in front:

$$\frac{du}{dx} = 4 \times 2 \cos(2x) = 8 \cos(2x)$$

3 Differentiate $v = \ln(8 - 4x^2)$ with the chain rule

$\ln()$ on the outside, $8 - 4x^2$ on the inside. Let $w = 8 - 4x^2$:

$$v = \ln w \Rightarrow \frac{dv}{dw} = \frac{1}{w} \quad w = 8 - 4x^2 \Rightarrow \frac{dw}{dx} = -8x$$

$$\frac{dv}{dx} = \frac{1}{8 - 4x^2} \times (-8x) = -\frac{8x}{8 - 4x^2}$$

4 Put it into the product rule

$$f'(x) = 8 \cos(2x) \cdot \ln(8 - 4x^2) + 4 \sin(2x) \cdot \left(-\frac{8x}{8 - 4x^2}\right)$$

$$f'(x) = 8 \cos(2x) \ln(8 - 4x^2) - \frac{32x \sin(2x)}{8 - 4x^2}$$

5 Tidy up (optional)

Divide the top and bottom of the fraction by 4 ($32 \div 4 = 8$ and $(8 - 4x^2) \div 4 = 2 - x^2$):

$$f'(x) = 8 \cos(2x) \ln(8 - 4x^2) - \frac{8x \sin(2x)}{2 - x^2}$$

Marks	Awarded for
1.5	$du/dx = 8 \cos(2x)$.
2	$dv/dx = -8x/(8 - 4x^2)$ using the chain rule.
2	Product rule applied correctly.
0.5	Final answer in a tidy form.

1(b)**[6 marks]**

Differentiate $h(t) = 7t^3 - \frac{5t^2}{2} + \frac{2}{t^4} + \sqrt[3]{t} - t + 1$.

ANSWER

$$h'(t) = 21t^2 - 5t - \frac{8}{t^5} + \frac{1}{3t^{2/3}} - 1$$

RULES YOU NEED

- Power rule: the derivative of t^n is $n \cdot t^{n-1}$, for any power n (negative and fractions too).
- $1/t^n = t^{-n}$ and $\sqrt[3]{t} = t^{1/3}$.
- A constant differentiates to 0.

WORKING — STEP BY STEP**1 Rewrite every term as a power of t**

The power rule only works on powers, so turn the fraction and the root into powers first:

$$\frac{2}{t^4} = 2t^{-4} \quad \sqrt[3]{t} = t^{1/3} \quad t = t^1$$

$$h(t) = 7t^3 - \frac{5}{2}t^2 + 2t^{-4} + t^{1/3} - t + 1$$

2 Differentiate term by term (power rule: bring the power down, take 1 off)

$$7t^3 \rightarrow 7 \times 3t^2 = 21t^2$$

$$\frac{5}{2}t^2 \rightarrow \frac{5}{2} \times 2t = 5t$$

$$2t^{-4} \rightarrow 2 \times (-4)t^{-5} = -8t^{-5}$$

$$t^{1/3} \rightarrow \frac{1}{3}t^{1/3-1} = \frac{1}{3}t^{-2/3}$$

$$t \rightarrow 1 \quad 1 \rightarrow 0 \text{ (a constant)}$$

$$(1/3 - 1 = 1/3 - 3/3 = -2/3.)$$

3 Put it together

Keep each sign from the original function:

$$h'(t) = 21t^2 - 5t - 8t^{-5} + \frac{1}{3}t^{-2/3} - 1$$

Written with positive powers:

$$h'(t) = 21t^2 - 5t - \frac{8}{t^5} + \frac{1}{3\sqrt[3]{t^2}} - 1$$

EXAM TECHNIQUE

Rewrite roots and fractions as powers **before** differentiating. The most common slip is $1/3 - 1$: it is $-2/3$, not $-1/3$.

Marks	Awarded for
1	$21t^2$.
1	$-5t$.
1.5	$-8t^{-5}$ (rewriting $2/t^4 = 2t^{-4}$).
1.5	$(1/3)t^{-2/3}$ (rewriting $\sqrt[3]{t} = t^{1/3}$).
1	-1 from $-t$, and 0 from the constant 1 .

1(c)**[5.5 marks]**Differentiate $g(\theta) = \frac{\sin^2(\theta)}{\cos(\theta)}$.**ANSWER**

$$g'(\theta) = \frac{2 \sin \theta \cos^2 \theta + \sin^3 \theta}{\cos^2 \theta} = \frac{\sin \theta (1 + \cos^2 \theta)}{\cos^2 \theta}$$

RULES YOU NEED

- Quotient rule: if $g = u/v$ then $g' = (u' \cdot v - u \cdot v') / v^2$.
- Derivatives: $\sin \theta \rightarrow \cos \theta$, $\cos \theta \rightarrow -\sin \theta$.
- $\sin^2 \theta + \cos^2 \theta = 1$.

WORKING — STEP BY STEP**1 Spot the structure: a fraction**

g is one function divided by another, so use the quotient rule:

$$u = \sin^2 \theta \quad v = \cos \theta$$

$$g'(\theta) = \frac{u' \cdot v - u \cdot v'}{v^2}$$

2 Differentiate $u = \sin^2 \theta$ with the chain rule

$\sin^2 \theta$ means $(\sin \theta)^2$. Outside: $(\)^2$, inside: $\sin \theta$. Let $w = \sin \theta$:

$$u = w^2 \Rightarrow \frac{du}{dw} = 2w \quad w = \sin \theta \Rightarrow \frac{dw}{d\theta} = \cos \theta$$

$$u' = 2 \sin \theta \cdot \cos \theta$$

3 Differentiate $v = \cos \theta$

$$v' = -\sin \theta$$

4 Substitute into the quotient rule

Watch the double minus: $-u \cdot v' = -\sin^2 \theta \cdot (-\sin \theta) = +\sin^3 \theta$:

$$g'(\theta) = \frac{(2 \sin \theta \cos \theta)(\cos \theta) - (\sin^2 \theta)(-\sin \theta)}{\cos^2 \theta} = \frac{2 \sin \theta \cos^2 \theta + \sin^3 \theta}{\cos^2 \theta}$$

5 Tidy up (optional)

Take out $\sin \theta$ from the top, then use $\sin^2 \theta + \cos^2 \theta = 1$ so that $2\cos^2 \theta + \sin^2 \theta = 1 + \cos^2 \theta$:

$$g'(\theta) = \frac{\sin \theta (2\cos^2 \theta + \sin^2 \theta)}{\cos^2 \theta} = \frac{\sin \theta (1 + \cos^2 \theta)}{\cos^2 \theta}$$

Marks	Awarded for
1.5	$u' = 2 \sin \theta \cos \theta$ (chain rule).
1	$v' = -\sin \theta$.
2	Quotient rule applied correctly.
1	Simplified answer (any correct equivalent form).

Question 1 Total: 17.5 Marks

Question 2

Critical points, optimisation — 17.5 marks

2(a)

[5.5 marks]

Find the critical points of $f(x) = x^3 - 9x^2 + 24x - 10$.

ANSWER

Critical points at $x = 2$ and $x = 4$, i.e. the points $(2, 10)$ and $(4, 6)$

RULES YOU NEED

- Critical point: where $f'(x) = 0$.
- Power rule: the derivative of x^n is $n \cdot x^{n-1}$.

WORKING — STEP BY STEP

1 What a critical point is

A critical point is where the slope is zero: $f'(x) = 0$. f is a polynomial, so f' exists everywhere.

2 Differentiate with the power rule

$$f'(x) = 3x^2 - 9(2x) + 24 - 0 = 3x^2 - 18x + 24$$

3 Solve $f'(x) = 0$

Every term has a factor of 3, so take it out. Then find two numbers that **multiply to 8** and **add to -6**: -2 and -4.

$$3(x^2 - 6x + 8) = 3(x - 2)(x - 4) = 0 \Rightarrow x = 2 \text{ or } x = 4$$

4 Find the y-values

$$f(2) = 8 - 9(4) + 24(2) - 10 = 8 - 36 + 48 - 10 = 10$$

$$f(4) = 64 - 9(16) + 24(4) - 10 = 64 - 144 + 96 - 10 = 6$$

5 Extra: what kind of points are they?

Not asked for, but the second derivative test shows it quickly: $f''(x) = 6x - 18$.

$$f''(2) = -6 < 0 \Rightarrow \text{local max at } (2, 10) \quad f''(4) = 6 > 0 \Rightarrow \text{local min at } (4, 6)$$

Marks	Awarded for
2	$f'(x) = 3x^2 - 18x + 24$.
2.5	Solved $f'(x) = 0$ to get $x = 2$ and $x = 4$.
1	Points $(2, 10)$ and $(4, 6)$.

2(b)**[12 marks]**

The surface area of a container in the shape of a right circular cylinder with no top is 3π m². What height h and base radius r will maximize the volume of the cylinder? Find this maximum volume. [Note: Total surface area = $\pi r^2 + 2\pi rh$, volume = $\pi r^2 h$.]

ANSWER

$r = 1$ m and $h = 1$ m, giving the maximum volume $V = \pi$ m³

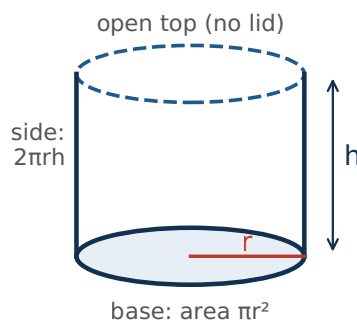
RULES YOU NEED

- Optimisation: write the quantity using one variable, differentiate, set the derivative to 0.
- Second derivative test: $f'' < 0$ at a critical point means a local maximum.

WORKING — STEP BY STEP**1 Write down the two formulas**

No top, so the area is the base plus the side (given in the question). The area must be 3π :

$$\pi r^2 + 2\pi rh = 3\pi \quad V = \pi r^2 h$$

**2 Use the area to write h in terms of r**

Divide every term by π , then rearrange for h :

$$r^2 + 2rh = 3 \Rightarrow 2rh = 3 - r^2 \Rightarrow h = \frac{3 - r^2}{2r}$$

3 Write the volume using r only

Substitute h into V and cancel one r from top and bottom:

$$V = \pi r^2 \cdot \frac{3 - r^2}{2r} = \frac{\pi r(3 - r^2)}{2} = \frac{\pi}{2}(3r - r^3)$$

r must be positive, and h must be positive too, so $3 - r^2 > 0$, i.e. $0 < r < \sqrt{3}$.

4 Differentiate and set $dV/dr = 0$

The $\pi/2$ stays in front; use the power rule on $3r - r^3$:

$$\frac{dV}{dr} = \frac{\pi}{2}(3 - 3r^2) = 0 \Rightarrow 3r^2 = 3 \Rightarrow r^2 = 1 \Rightarrow r = 1$$

($r = -1$ is rejected because a radius must be positive.)

5 Check it really is a maximum

Second derivative test:

$$\frac{d^2V}{dr^2} = \frac{\pi}{2}(-6r) = -3\pi r \Rightarrow \text{at } r = 1: -3\pi < 0 \Rightarrow \text{maximum}$$

(Also, V is 0 at both ends $r = 0$ and $r = \sqrt{3}$, and positive in between, so the one critical point must be the maximum.)

6 Find h and the maximum volume

$$h = \frac{3 - 1^2}{2 \times 1} = \frac{2}{2} = 1 \text{ m}$$

$$V = \pi \times 1^2 \times 1 = \pi \text{ m}^3$$

EXAM TECHNIQUE

Always use the extra condition (here the area) to get rid of one variable before differentiating. You can't maximise $V = \pi r^2 h$ while it still has two letters.

Marks	Awarded for
3	Using the area 3π to get $h = (3 - r^2)/(2r)$.
2.5	$V(r) = (\pi/2)(3r - r^3)$.
2.5	$dV/dr = (\pi/2)(3 - 3r^2) = 0$, giving $r = 1$.
1	$h = 1 \text{ m}$.
1.5	Check that this is a maximum (second derivative or end values).
1.5	Maximum volume $V = \pi \text{ m}^3$.

Question 2 Total: 17.5 Marks

Question 3

Representing functions, related rates of change — 17.5 marks

3(a)

[5.5 marks]

State four ways to represent a function.

ANSWER

1. By a formula (algebraically). 2. By a table of values (numerically). 3. By a graph (visually). 4. In words (verbally).

Note on the lecture material. The MA1001 notes don't list "four ways to represent a function" as such. There is no introductory functions lecture in the combined notes. This answer uses the standard four (formula, table, graph, words), each of which appears in the notes: formulas throughout, tables of values and graphs in graph sketching, and word statements in physical laws.

WORKING — STEP BY STEP

1 By a formula (algebraically)

A rule that says what to do to the input, for example:

$$f(x) = x^2$$

2 By a table of values (numerically)

A list of inputs and the outputs they give, as used for graph sketching in the lectures:

x	0	1	2	3
f(x) = x²	0	1	4	9

3 By a graph (visually)

All the points $(x, f(x))$ plotted in the plane, e.g. the U-shaped parabola $y = x^2$. Each vertical line meets the graph at most once.

4 In words (verbally)

A description in a sentence, for example Charles's Law from the lectures: "the volume of a gas is proportional to its temperature" ($V = CT$).

EXAM TECHNIQUE

An arrow or “machine” diagram ($x \rightarrow f \rightarrow f(x)$), as used for composition in the lectures, is also a valid way to show a function.

Marks	Awarded for
1	Formula / algebraic.
1	Table of values / numerical.
1	Graph / visual.
1	Words / verbal.
1.5	A suitable example for the representations.

3(b)**[12 marks]**

The volume of a sphere is increasing at the rate of $24\pi \text{ cm}^3$ per second. Find the rate of increase of the radius when the radius is 6 cm.

ANSWER

$dr/dt = 1/6 \text{ cm per second}$

RULES YOU NEED

- Sphere: $V = (4/3)\pi r^3$ (formula sheet).
- Related rates: differentiate the formula with respect to time t ; $d/dt(r^3) = 3r^2 \cdot dr/dt$.

WORKING — STEP BY STEP**1 Write down the formula linking V and r**

From the formula sheet (sphere):

$$V = \frac{4}{3}\pi r^3$$

2 Write down what we know and what we want

$$\frac{dV}{dt} = 24\pi \text{ cm}^3/\text{s} \quad \text{when} \quad r = 6 \text{ cm}$$

We want $\frac{dr}{dt}$, how fast the radius is growing.

3 Differentiate both sides with respect to time t

r changes with time, so the chain rule adds dr/dt . The power rule on r^3 gives $3r^2$, and $4/3 \times 3 = 4$:

$$\frac{dV}{dt} = \frac{4}{3}\pi \times 3r^2 \times \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

4 Put in the numbers

$r = 6$, so $r^2 = 36$ and $4\pi \times 36 = 144\pi$:

$$24\pi = 4\pi(6)^2 \frac{dr}{dt} = 144\pi \frac{dr}{dt}$$

5 Solve for dr/dt

Divide both sides by 144π . The π cancels, and $24/144 = 1/6$ (both divide by 24):

$$\frac{dr}{dt} = \frac{24\pi}{144\pi} = \frac{1}{6} \text{ cm/s}$$

Marks	Awarded for
2	$V = (4/3)\pi r^3$.
4	$dV/dt = 4\pi r^2 dr/dt$ (chain rule with respect to t).
3	Substituting $dV/dt = 24\pi$ and $r = 6$.
3	$dr/dt = 1/6$ cm/s.

Question 3 Total: 17.5 Marks

Question 4

Rates of change: velocity and acceleration — 17.5 marks

Given: $s(t) = 10 + 27t - t^3$ (metres, t in seconds)

4(a)(i)

[6 marks]

Find the speed of the body after 2 seconds.

ANSWER

Speed = 15 m/s

RULES YOU NEED

- Velocity $v = ds/dt$; acceleration $a = dv/dt$.

WORKING — STEP BY STEP

1 Velocity is the derivative of displacement

Velocity v is the rate of change of displacement s with respect to time t :

$$v = \frac{ds}{dt}$$

2 Differentiate $s(t) = 10 + 27t - t^3$

The constant 10 gives 0, $27t$ gives 27, and t^3 gives $3t^2$:

$$v(t) = 0 + 27 - 3t^2 = 27 - 3t^2$$

3 Put in $t = 2$

$$v(2) = 27 - 3(2)^2 = 27 - 3 \times 4 = 27 - 12 = 15 \text{ m/s}$$

Speed is the size of the velocity, so the speed is **15 m/s**.

Marks	Awarded for
3	$v = ds/dt$.
1.5	$v(t) = 27 - 3t^2$.
1.5	$v(2) = 15 \text{ m/s}$.

4(a)(ii)**[6 marks]**

After how many seconds is the body at rest?

ANSWER **$t = 3$ seconds****WORKING — STEP BY STEP****1 “At rest” means the velocity is zero**

$$v(t) = 27 - 3t^2 = 0$$

2 Solve for t Add $3t^2$ to both sides, then divide by 3:

$$3t^2 = 27 \Rightarrow t^2 = 9 \Rightarrow t = 3 \text{ or } t = -3$$

3 Reject the negative timeTime is measured forwards from the start, so $t \geq 0$. The body is at rest after **3 seconds**.Check: $v(3) = 27 - 3(9) = 27 - 27 = 0 \checkmark$

Marks	Awarded for
2	Setting $v = 0$.
2	$t^2 = 9$.
2	$t = 3$ s ($t = -3$ rejected).

4(a)(iii)**[5.5 marks]**

Find the acceleration of the body after 2 seconds.

ANSWER

$$a = -12 \text{ m/s}^2 \text{ (the body is slowing down)}$$

WORKING — STEP BY STEP**1 Acceleration is the derivative of velocity**

$$a = \frac{dv}{dt}$$

2 Differentiate $v(t) = 27 - 3t^2$

$$a(t) = 0 - 3 \times 2t = -6t$$

3 Put in $t = 2$

$$a(2) = -6 \times 2 = -12 \text{ m/s}^2$$

The minus sign means the acceleration is against the direction of motion: the body (moving at +15 m/s) is slowing down.

Marks	Awarded for
3	$a = dv/dt = -6t$.
2.5	$a(2) = -12 \text{ m/s}^2$.

Question 4 Total: 17.5 Marks

Question 5

Tangent lines, implicit differentiation — 17.5 marks

5(a)

[5.5 marks]

At what point on the curve $y = e^x$ is the tangent line parallel to the line $y = 2x$?

ANSWER

At the point $(\ln 2, 2)$

RULES YOU NEED

- Parallel lines have equal slopes.
- The derivative of e^x is e^x .
- $\ln(e^x) = x$ and $e^{\ln y} = y$.

WORKING — STEP BY STEP

1 Parallel lines have the same slope

$y = 2x$ is in the form $y = mx + b$ with $m = 2$. So we need the tangent to $y = e^x$ to have slope 2.

2 The slope of the tangent is the derivative

The derivative of e^x is e^x itself:

$$\frac{dy}{dx} = e^x$$

3 Set the slope equal to 2 and solve

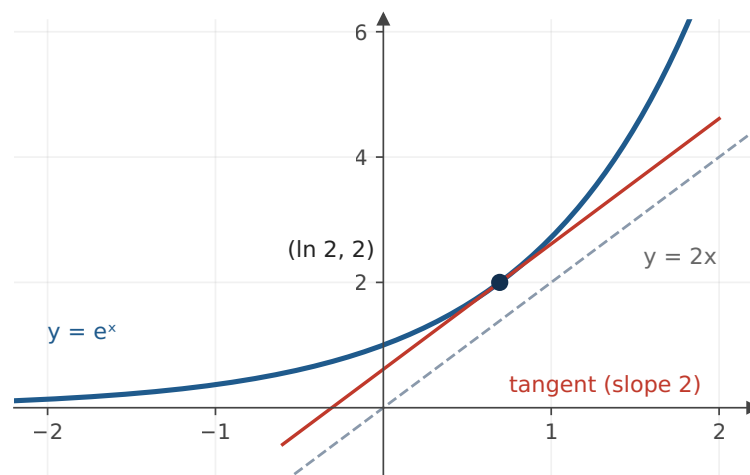
Take $\ln$ of both sides ($\ln$ undoes e : $\ln(e^x) = x$):

$$e^x = 2 \Rightarrow x = \ln 2$$

4 Find the y-coordinate

e and $\ln$ cancel each other out: $e^{\ln 2} = 2$.

$$y = e^{\ln 2} = 2 \Rightarrow \text{point } (\ln 2, 2)$$



Marks	Awarded for
1	Slope of $y = 2x$ is 2.
1.5	$dy/dx = e^x$.
2	$e^x = 2$, so $x = \ln 2$.
1	Point $(\ln 2, 2)$.

5(b)**[12 marks]**

Find the slope, and hence the equation, of the tangent line to the graph of $y(x)$ at the point $p = (1, 1)$ where $y^4 + xy = x^3 - x + 2$.

ANSWER

Slope $dy/dx = 1/5$; tangent line $y = \frac{1}{5}x + \frac{4}{5}$

RULES YOU NEED

- Implicit differentiation: differentiate both sides with respect to x ; $d/dx(y^n) = n \cdot y^{n-1} \cdot dy/dx$.
- Product rule for xy : $d/dx(xy) = y + x \cdot dy/dx$.
- Tangent line: $y - b = m(x - a)$.

WORKING — STEP BY STEP**1 Check the point is on the curve**

$$\text{Left: } 1^4 + 1 \times 1 = 2 \quad \text{Right: } 1^3 - 1 + 2 = 2 \quad \checkmark$$

2 Use implicit differentiation

We can't easily get y on its own, so differentiate both sides with respect to x , remembering that y is a function of x . Every time we differentiate something with y in it, the chain rule adds a factor dy/dx .

3 Differentiate y^4

Power rule, then the chain rule adds dy/dx :

$$\frac{d}{dx}(y^4) = 4y^3 \frac{dy}{dx}$$

4 Differentiate xy with the product rule

$u = x$ (so $u' = 1$) and $v = y$ (so $v' = dy/dx$):

$$\frac{d}{dx}(xy) = 1 \cdot y + x \cdot \frac{dy}{dx} = y + x \frac{dy}{dx}$$

5 Differentiate the right side

$$\frac{d}{dx}(x^3 - x + 2) = 3x^2 - 1$$

6 Put it together and substitute $x = 1, y = 1$

$$4y^3 \frac{dy}{dx} + y + x \frac{dy}{dx} = 3x^2 - 1$$

$$4(1)^3 \frac{dy}{dx} + 1 + 1 \cdot \frac{dy}{dx} = 3(1)^2 - 1 \Rightarrow 4 \frac{dy}{dx} + 1 + \frac{dy}{dx} = 2$$

7 Solve for the slope

Collect the dy/dx terms ($4 + 1 = 5$) and subtract 1 from both sides:

$$5 \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{5}$$

8 Write down the tangent line

Point (1, 1), slope 1/5, using $y - b = m(x - a)$:

$$y - 1 = \frac{1}{5}(x - 1) \Rightarrow y = \frac{1}{5}x - \frac{1}{5} + 1 = \frac{1}{5}x + \frac{4}{5}$$

(Multiplying by 5 gives the equivalent form $x - 5y + 4 = 0$.)

EXAM TECHNIQUE

Substitute $x = 1$ and $y = 1$ straight after differentiating. It is much easier than rearranging for dy/dx in general first.

Marks	Awarded for
1	Point (1, 1) checked on the curve.
2.5	$d/dx(y^4) = 4y^3 dy/dx$.
2.5	$d/dx(xy) = y + x dy/dx$ (product rule).
1	Right side $3x^2 - 1$.
3	Substituting (1, 1) and solving: $dy/dx = 1/5$.
2	Tangent line $y = x/5 + 4/5$.

Question 5 Total: 17.5 Marks